Sailab Banik

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SUMMARY

Software Engineer with 3 years of production experience, the last 2 building LLM-powered document intelligence systems: a platform used by **20+ business teams**, and vision-model extraction from scanned engineering drawings. Specializes in **Python, FastAPI, React, RAG pipelines, and LLM validation**, with end-to-end ownership from architecture to deployment.

WORK EXPERIENCE

Software Engineer II Oct 2024 – Present
ZF Group *Chennai, India*

- Developed the AI layer of **PrimeAI**, an internal document processing platform adopted by **20+ business teams**, covering extraction, user-defined models, and retrieval; reduced manual review by **60%**, saving **2,000+ hours per quarter**.
- Routed extraction between **locally served models on Ollama**, for documents that cannot leave internal infrastructure, and frontier APIs where accuracy required them, closing most of the gap with retrieval rather than a larger model.
- Architected a drawing extraction pipeline on **Azure OpenAI** for **3,000+ scanned engineering drawings**, with multi-model validation and an **evaluation harness** tracking cost and coverage per prompt version.
- Owned product flows end to end as frontend lead, from UI to API and database; designed **role- and group-based access control**, enabling rollout to departments with strict data-access requirements.
- Drove CI/CD improvements for the team, shortening release cycles by **40%** and lowering production hotfixes by **35%**; introduced frontend test suites with **React Testing Library and Vitest**, trimming manual testing by **60%**.

Software Engineer (Graduate Trainee) Aug 2023 – Sep 2024
ZF Group *Chennai, India*

- Resolved **10+ high-priority production bugs** on a business-critical legacy platform, cutting user-reported issues by **45%**.
- Led the **security hardening pass** for internal applications ahead of audit, remediating injection and input-validation findings and carrying every application through review.
- Optimized operations scheduling workflows, improving throughput by **30%**.

PROJECTS

ContextDesk Lab: Context Engineering Playground for Redis Iris *Python, FastAPI, Redis Iris, Next.js, TypeScript, Docker*

- Cut repeat-question latency from **4.65s to 917ms with 0 tokens billed** by answering semantically equivalent prompts from **LangCache**; cache misses still report their similarity score.
- Designed an AI support console that returns an execution record with every response, orchestrating **Redis Iris LangCache, Agent Memory, and Context Retriever**, with **50 offline tests** over stubbed components.

FocusTrack: Work Tracker with AI Reviews | **focustrack.sailab.dev** *Next.js, TypeScript, Supabase, PostgreSQL, Vercel*

- Shipped a tracker that splits work sessions into execution and preparation, with daily, weekly, and monthly **AI reports** citing those numbers; a database claim caps provider calls at one per user per period.
- Configured **Supabase** Postgres and Auth with **row-level security** on every user table, verified by **60 pgTAP tests**; set up DNS and a GitHub Actions pipeline gating each **Vercel** deploy.

QueueLens: Distributed Job Queue with Live Dashboard *Python, FastAPI, Redis, PostgreSQL, Next.js, Docker*

- Built a producer–consumer job queue: a **FastAPI** producer persists jobs in **PostgreSQL** and hands job IDs to background workers over a **Redis** list (LPUSH/BRPOP), with a **Next.js** dashboard tracing every job through its lifecycle live.
- Implemented **retries with exponential backoff**, Redis-backed **per-client rate limiting** shared by all API processes, and a queue cap returning 429/503 with Retry-After; ordered state commits before enqueueing to prevent API–worker race conditions.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML, CSS
AI/ML: Generative AI, RAG Pipelines, Large Language Models (LLMs), Vision-Language Models (VLMs), Structured Outputs, LLM Evaluation, Semantic Caching, Context Engineering, Azure OpenAI, Ollama, LangChain, Redis Iris (LangCache, Agent Memory)
Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, Supabase (Auth, RLS), Redis, OAuth2 / SSO (Microsoft Entra ID, MSAL)
Frontend: React.js, Next.js, TypeScript, Redux Toolkit, Tailwind CSS, Material UI, React Testing Library, Vitest, MSW
DevOps & Tools: Docker, Docker Compose, CI/CD Pipelines, GitHub Actions, Azure DevOps, Vercel, DNS, Git

EDUCATION

ICFAI University Tripura Aug 2019 – Jul 2023
*B.Tech in Computer Science and Engineering – **CGPA: 9.64*** *Agartala, India*

AWARDS & CERTIFICATIONS

Budding Star Award — ZF Group, Recognition for Business Impact in the first year *2024*
System Design Cohort — ChaiCode: distributed systems, load balancing, caching, sharding, queues Credential | *Jun 2026*